

Group Assignment Instructions

1. Each team must choose one problem from the provided list (each team must work on a different problem).
2. Solve the problem using machine learning and the given dataset (using scikit-learn or other relevant libraries).
3. Try at least 2 different machine learning algorithms and compare their performance.
4. Prepare a slide deck and deliver a 15-minute presentation during the next session.

Presentation Requirements

1. Team & Project Setup

- Introduce your team members and describe how you split the workload.
- Describe the problem: What real-world task were you trying to solve, and what was your target variable? What type of problem is it? (regression, binary classification, multi-class classification, etc.)

2. Data Exploration & Preprocessing

- Summarise the dataset: What were the key features, total size, and train/test split ratio?
- Mention at least one surprise, challenge, or dirty data issue you found during initial data exploration and how you handled it.

3. Model Selection & Comparison

- Briefly explain the 2+ algorithms you chose (plain language, no math).
- How do these models compare against a simple baseline (e.g. always predicting the most frequent class)?

4. Evaluation & Error Analysis

- Report your evaluation metrics (Accuracy, Precision, Recall, F1, etc.) and present a confusion matrix for your best model.
- Failure Analysis: Show 2 specific examples from the test set that your model got wrong. Why do you think it made those mistakes?
- Which features turned out to be the most and least useful?

5. Reflection & Practical Viability

- Did your models show signs of overfitting or underfitting? How do you know?
- If you had more time or additional resources, what new features or improvements would you add?
- Real-World Verdict: Is your model reliable enough to be deployed in a real-world scenario today? Why or why not?

List of challenges (choose one)

Challenge 1: Poisonous mushrooms

Determine if a mushroom is poisonous or not.

Information and dataset: <https://www.kaggle.com/datasets/uciml/mushroom-classification>

Challenge 2: Personality prediction

Predict a person's personality type.

Information and dataset: <https://www.kaggle.com/datasets/miadul/introvert-extrovert-and-ambivert-classification>

Challenge 3: Twitter sentiment analysis

Determine the sentiment of Twitter posts.

Information and dataset: <https://www.kaggle.com/datasets/mdismielhossenabir/sentiment-analysis>

Challenge 4: Loan approval

Determine if a client's loan application will be approved.

Information and dataset: <https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data>

Challenge 5: Star types

Classify stars.

Information and dataset: <https://www.kaggle.com/datasets/deepu1109/star-dataset>

Challenge 6: Crop recommendation

Recommend the most suitable crop for a farm.

Information and dataset: <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>